**CP/M is one of the most widely used operating systems, particularly in the business area. Any improvement in its performance or ease of use will benefit a wide range of readers. The CP/M Card from Advanced Logic Systems offers a Z80 card together with the new CP/M Plus operating system and other utilities all in one package for £300.**

**Several commentators agree that it has the potential to be an excellent product. But as TOM DERWENT says in this article, some people have reported some early teething problems that might trouble an inexperienced user. An Apple dealer in Germany had similar difficulties. We publish his views here, together with a reply from the manufacturers.**

# A versatile CP/M Card – but take care

By TOM DERWENT

I WAS very pleasantly surprised when I fired up CP/M Plus. It contains all the usual CP/M utilities, some public domain software, such as CAT, but with a Digital Research copyright message and some new ones such as the symbolic instruction debugger, an upgraded version of DDT, a good Z80 macro assembler and a linker, among others.

The software comes on three discs, but only the first one contains a bootable system. The other two contain the utilities and the CBasic programming language.

Curiously, disc No 1 has a write-enable notch, and will not boot if it is covered up. The others do not have the notch and so are permanently write protected. It is important to copy these master discs, and file the originals in a safe place.

The manual is impressive, if not daunting. It is a two inch thick, cloth bound, book shaped ring binder containing three separate publications. The outer box can act as a storage container when placed on a bookshelf.

Luckily, it is not necessary to digest all of this material before using the CP/M Card. The manuals are the CP/M Card User's Guide, a 40 page offering by ALS; CP/M Plus (CP/M version 3) Operating System User's Guide (½in thick); and CBasic Language Reference Manual (slightly thinner), both from Digital Research.

The first 16 pages of the ALS book cover installation of the card and the fundamentals of the CP/M 3 operating system – concluding with a section relating some CP/M 3 commands to DOS equivalents – enough to get you started.

The rest of this book comprises a description of the utilities supplied (eight pages), a hardware description (four pages) for the technically minded, and an eight page glossary of terms which would be extremely useful for the novice.

The two manuals from Digital Research are much more in the reference manual line and as reference manuals go are quite good, with meaningful headings in the table of contents and lots of examples in red print, often showing exactly what would appear on the screen, plus indices.

However, one should realise that they are reference manuals and treat them accordingly. That is, they are excellent for looking up the required form of a command or CBasic statement, but you should not try to learn programming from them. It will probably be necessary for some users to buy one of the many CP/M operating system guides by independent authors.

Notwithstanding the above, the documentation provided should certainly prove sufficient for those users who merely wish to run pre-packaged programs and for enthusiasts who already have some computing knowledge.

It is essential after reading at least the first 16 pages of the ALS User's Guide to back up the discs using the relevant option on the start-up menu automatically presented when the system is booted.

This option takes you into the DSKCOPY utility which presents a further menu, including simple formatting, and copying of system tracks, as well as a whole disc copy option which will automatically format the destination disc.

However DSKCOPY always copies from drive A to B, so if you only have one drive you have a problem. You would have to obtain a copy program – either from public domain CP/M-80 software, buy a proprietary one, or else write one yourself.

In my view DSKCOPY is the most useful of the utilities provided. The others are:

- ADUMP – will give hexadecimal and Ascii dumps of files or sectors on a disc.
- MON 65 – permits programmers access to the Apple 6502 monitor.
- APPMAKER – creates discs containing the basic CP/M 3 system files, to which application packages may be added.
- HELLO – the program executed on startup to provide the initial "CP/M CARD" copyright notice and menu. It is not necessary to the system and one of its options is to remove the menu. In fact the HELLO.COM file may be removed and PROFILE.SUB modified or removed, to give additional free disc space.
- WSMAKER – a program which will automatically install Wordstar. This program is a victim of progress. I have successfully installed Wordstar 3.01P using this program, but the current one on sale is Wordstar 3.3, on which it will not work. In any case I do not fancy all the options chosen for me by this program and it cannot install specialist printers, only what MicroPro term "standard printer".

I recommend users to install their own, or have their local dealer install it for them. The UK distributors of the ALS CP/M Card, Scope Systems, on request provide a three page guide to be read in conjunction with the installation manual supplied with Wordstar. This does assume that the user will look up commands in the reference manual.

I have installed and used Wordstar using the ALS Smarterm II 80 column board, the Apple 80 column board, the Apple extended memory 80 column board, and a Vision-80 board, though I have not worked out how to get the inverse-video around the command option block as is done by WSMaker or WS 3.01P. Two other users have said they could install Wordstar – but the characters that should be in INVERSE are normal and vice versa.

The CP/M itself is far superior to CP/M 2.2 and, with the different processor, faster in operation than its older brother.

I have always been impressed with the very high standard of software that is available under CP/M, but the operating system has left a lot to be desired. CP/M Plus goes a long way to improving its image.

I spent a few happy hours playing with the many new commands and transient

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programs that come with the package. I haven't listed them, or gone into any great detail as this would take a complete article on its own.

Instead I will pick a few facilities from which I can see an obvious benefit. Firstly, SUBMIT files. These are files which contain other CP/M 3 commands and hence can be used to create job streams or batch runs.

A special case of this is the PROFILE.SUB which is resident on system discs and is searched for and executed on startup if present. The HELLO program is called from PROFILE.SUB. This facility enables the operating environment (for example, the disc/file type search order using SETDEF, or the date/time set routine) to be set up automatically for an application disc. Further, the applications program could be called from the .SUB file – all without user action after the cold start.

The USER option allows files to be stamped with a user number from 0-16 and such files do not appear on other users catalog listings – could be very useful in a hard disc situation with many files. SHOW and CAT replace STAT command in giving information about disc contents, types, free space, etc. DIR remains, but is not as useful as CAT.

SET sets file, drive attributes, volume labels, password protection functions and date and time file stamping options. Program developers could use the date/time stamping to avoid losing track of which was the latest version of a file, as could other users who have time dependent files such as accounts files.

There are many other commands in CP/M Plus which have enhanced features – I have not even mentioned those to do with the assembly language programming facilities. Possibly this aspect could be dealt with in a separate article. Suffice it to say that this is CP/M Plus, not a variant of CP/M version 2.

I accomplished the installation within half an hour, including reading time – the card will go in any slot and still boot up (I have tried it in slots 1 to 7 on an Apple IIe, though when in slot 6, I had to move the disc controller to slot 7). Note that for an Apple II, a 16k RAM expansion card is necessary. ALS recommend slots 4 or 7, to fit in with other conventional slot selections.

For the inexperienced user it is possible that CP/M may be a bit much to swallow all at once, but it is possible to start small and build one's knowledge. The small computer system cannot devote the same amount of resources to the user interface as larger systems can.

Most CP/M 2.2 applications programs should work immediately under CP/M 3 – some problems may be encountered with those which have been written to intimately fit or interact with a previous operating system. Best procedure is to try before you buy at your local dealer.

The lack of hard disc support at the present time is disappointing. However the BIOS section has to be modified for every new hardware item and individual hardware manufacturers should produce their own drivers within a few months.

I am told that one networking firm has developed their own BIOS and can now take advantage of the 32 megabyte file sizes possible under CP/M 3.

The ALS manual mentions a GSX-80 graphics extension programmer's guide. This is not available because DRI have changed their approach to graphics – at least for Apple CP/M Plus. Graphics will be supplied as an extension to CBasic at some time in the future. As CP/M is largely a business operating system, perhaps that is not too important. Assembly language games should still work, but much faster.,

The CBasic supplied in the package is not the native code compiler version, but the one which translates source code into intermediate code which is executed by a run-time interpreter.

Overall, what is provided is something for everyone – a 6 mHz Z-80B processor for scientific and technical number crunching, a sophisticated operating system for enthusiasts and programmers, facilities for high and low level programming, and a cheap CP/M machine for businessmen.

An Apple owner with a CP/M Card would have access to the two largest existing program collections in the world, Apple DOS and CP/M, as well as the future developments which the advanced features of CP/M Plus will make possible.

However, as with any new item I would advise prospective customers who are upgrading their systems to ensure that the card works with their present system and software before they buy it. This can only be done by trying it, whether at the dealers', or at your own place of business.

**Windfall has received the following letter from a dealer in Germany, Karl-Heinz Weiss of Wilhelmshaven.**

I am a dealer and like to sell products, but with this card I am still cautious. The card has many good features, the hardware is all right, but the finished product was not finished or satisfactory at the time of writing (August 20) in terms of software.

The GSX-80 part is still missing although it was announced to be ready for shipment in the autumn. You cannot use any printer interface that works like the Grappler card, for example Microbuffer or Microtek's dumpling 64.

You cannot use drives other than the Apple disc drives, for example 8 inch with a controller like VISTA A800, unless you write all the necessary code yourself.

Several RAM cards like Saturn's or Vergecourt's Ramex cannot be installed with the CP/M Card for use as a RAM disc. If you have a normal Apple keyboard and a Videx Videoterm board or similar brands, you encounter trouble when you use the CP/M card with Wordstar or Datastar.

The WSMaker utility on the disc supplied by ALS does configure Wordstar, but only for use with ALS' own 80 column board – Smartterm.

However, if you use some kind of keyboard with separate keys for CONTROL and left square bracket ] you will have no trouble at all.

You cannot use the MBasic supplied with the original Softcard and put it to use with the CP/M card. Instead you must take the Lifeboat Basic-80. At the moment the CP/M card has no utility like CONFIGIO to change character definitions.

Another important point is that the difference between free space with the Softcard's MBasic on the one hand and the Basic-80 and the CP/M Plus Card on the other is slightly more than 11 K1<.

● **This reply is from Bob Ackerman, director of sales, Advanced Logic Systems.**

*GSX-80, as of this writing, has not been delivered by Digital Research, and therefore has not been delivered to any CP/M Card customers.*

*After delivery of a number of cards, we found that there were some shortcomings in the printer driver. Revision B software has now been released, solves the problem fully, and various customers indicate full satisfaction when using the wide variety of cards in the market.*

*The current release software does indeed address only the Apple controller. We are unable to define the full set of controller drivers that will satisfy everyone, so have been providing CP/M Cards to the controller developers. We expect that there will be a number of alternate controllers supported shortly by the manufacturers of the boards.*

*RAM discs fit into the same category as other controllers, and require drivers, dependent on hardware. We cannot select the card that will satisfy everyone, so intend to support the various card manufacturers.*

*The comment made about WSMaker is correct with respect to Revision A only. Revision B replaces that program, recognising the Apple IIe text card, Videoterm and its copies, and also Smarterm boards.*

*MBasic is a version of Basic-80 designed by Microsoft for specific application to the softcard operating system only. It does work on the CP/M Card, but may not return to CCP properly.*

*The softcard is the only CP/M implementation that we know of that provides a program like CONFIGIO, written to work with MBasic to convert certain keyboard characters on the Apple II.*

*ALS's CP/M 2.2 and CP/M Plus take the more standard approach of converting none of the keyboard input characters.*

*I am aware that there are Z-80 cards sold without software, requiring either purchase of CP/M and the generation of a BIOS or the copying of someone else's software. One case is expensive and inefficient. The other is illegal in my country and unethical in most.*

*Regarding free space, the transient program area of the CP/M Card is 60.5k, we believe the largest currently available. Supercalc signs on showing 35k of model space on the CP/M Card, and only 25k on the CP/M 2.2 implementations.*

*We regret that Mr Weiss had difficulty with the card, and intend to continue to upgrade our products and support.*